Zilionis 2019 Non-Small Cell Lung Cancer — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

6/7 samples (≥132 cells) intermix on one continuous manifold with no per-patient islands. An integration sweep (θ 0→10) read on the embedding gives one connected kNN component at every θ (iLISI 1.6→3.0 of 6); mixing plateaus at θ=10. Mixed histology (LUSC p1 + LUAD p3–p7) is interspersed (LUSC kNN self-enrichment 2.3×, far below an island); manifold roundish (aspect 1.1).

Gate 2 — module geography → GO (non-branching, Crest-flagged)

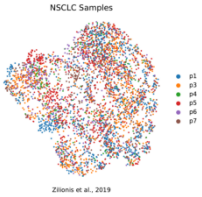
Three embryonic programs are present and MUTUALLY EXCLUSIVE at single-cell resolution (raw Gast~Crest −0.47, pSom~Crest −0.48; all smoothed cross-repressions p<0.001 vs a permutation null — REAL, not a smoothing artifact). No Gast/pSom co-expression hub (Gast&pSom-high enrichment 0.39) → no bifurcation. Along pseudotime Gast declines (ρ=−0.44) and pSom rises (ρ=+0.48). Crest FLAGGED: a real anti-correlated territory but U-shaped/non-terminal (ρ=+0.01, p=0.86; quadratic-U p<0.001) — departs from the reference terminal-Crest readout.

Gate 3 — hub mutual exclusivity → NA

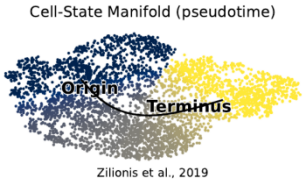
Single non-branching axis — the curve is 2 tips / 0 forks and fold-free (self-approach 1.68), tree fits over-fragment (5 tips/3 forks, unstable across λ). No Gast/pSom-overlap hub, so no branch pair to test.

STEP FIGURES

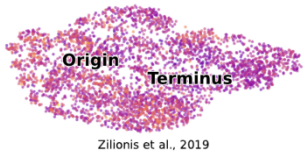
Gate 1: integration



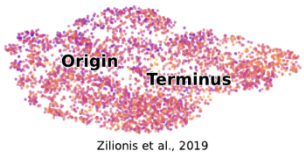
Trajectory / root



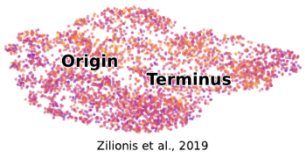
Gast



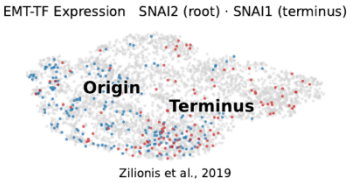
pSom



Crest



EMT-TFs



Dataset. NSCLC (Zilionis et al., 2019), inDrop — 7 samples / 31,179 cells (master scope); malignant only: 3,995 cells; after ≥132-cell thresholding 3,920 across 6 tumour-bearing samples; island removal 3,920 (one connected component, 0 removed) enter the trajectory (whole manifold, no scoping).

Verdict. ✓ Non-branching pattern reproduces with a Crest flag — Gate 1 GO, Gate 2 GO, Gate 3 NA. A single monotone Gast→pSom developmental axis (gastrulation-high origin → somitogenesis-high terminus); Crest is a real but non-terminal, off-axis third territory, so the manifold lacks Puram's hub-and-spoke bifurcation and Gate 3 has no hub to test. The three modules are cell-cycle-free (Gast 0/81, pSom 2/98, Crest 0/104 cell-cycle genes), cycling is low and diffuse (6.4%), and the Gast~pSom anti-correlation survives in non-cycling cells (−0.33→−0.33) — the axis is developmental, not proliferative. EMT-TFs are weak/early-loaded: SNAI2 (root) / SNAI1 (terminus) by pseudotime, FOXC1 the best-expressed factor (recovered by widening to FOXC/PRRX). Stochastic note: Harmony KMeans is not seed-deterministic; counts / iLISI vary ±~1% between runs (benign, not a changed call).

Parameter choices

Min cells/sample	132 geom-mean (cliff 232→p2 75; 6/7)	**Integration**	Harmony θ=10; 20 PCs; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip = Gast/pSom-high, Crest-low	**Poles**	Origin ~2,070 / Terminus ~1,850